

Tunneling, and What Makes Quantum Information Different

15, 17 September · Objective 2

Tunneling is the clearest case of a quantum effect with unambiguous engineering consequences — it powers flash memory, scanning tunneling microscopes, and stellar fusion, and it sets a hard floor on transistor scaling. Then the shift from physics to information: what a qubit is as a *resource*, and why no-cloning is the theorem your entire professional intuition should recognize.

Tunneling

A particle with energy E meets a barrier of height $V > E$. Classically the region is forbidden — kinetic energy would be negative. Quantum mechanically, ψ does not stop at the wall; it decays exponentially inside it, and if the barrier is thin, a nonzero amplitude emerges:

Inside barrier:

$$\psi(x) \propto e^{-\kappa x}$$

$$\kappa = \sqrt{2m(V-E)}/\hbar$$

$$T \approx e^{-2\kappa d}$$

The transmission probability is exponential in barrier width and in $\sqrt{m}$. Two consequences worth internalizing:

- **Exponential sensitivity.** An STM exploits this: tunneling current changes by ~an order of magnitude per ångström of tip height, which is why it resolves individual atoms.
- **Mass dependence.** Electrons tunnel readily; protons rarely; anything macroscopic never. The $\sqrt{m}$ in κ is why this is a microscopic phenomenon.

Applications: flash memory writes by forcing electrons through an oxide barrier into a floating gate. Alpha decay is tunneling out of a nuclear potential — Gamow's 1928 model explained the Geiger-Nuttall relation, spanning 20 orders of magnitude in half-life. Stellar fusion requires tunneling through Coulomb repulsion; without it, the Sun does not burn.

And the constraint: below roughly 5 nm of gate oxide, leakage current from tunneling becomes unmanageable. This is a principal reason Dennard scaling ended — a quantum effect setting the economics of the semiconductor industry.

Analogy boundary — tunneling as a rare event. Treating it as a low-probability path through a rejected branch gets the statistics right and the mechanism wrong. Your rare event is a real trajectory taken infrequently. There is no trajectory through the barrier — the particle is never *in* the forbidden region with negative kinetic energy. Asking "how long did it take to cross" has no settled answer (the tunneling-time problem is still live). The transmitted amplitude is a boundary-matching result, not a path.

Bits and qubits

The comparison, stated as an interface:

CLASSICAL BIT	QUBIT
State: $b \in \{0, 1\}$	State: $\alpha 0\rangle + \beta 1\rangle$, $\alpha, \beta \in \mathbb{C}$
2 states	Continuum (surface of a sphere)
Read: returns b , non-destructive	Read: returns 0 or 1 probabilistically, destroys the superposition
Copy: trivial (fan-out)	Copy: impossible (no-cloning)
n bits: n values	n qubits: 2^n amplitudes
Operations: any function	Operations: unitary only (reversible)

The row that misleads: n qubits require 2^n complex amplitudes to describe, and this is routinely reported as "exponential parallel computation." It is not. You can only **read out n bits**. Measurement collapses the 2^n amplitudes to one n -bit string. The exponential description is real; exponential *access* is not.

Quantum advantage therefore never comes from parallelism alone. It comes from arranging interference so that wrong answers cancel and right answers reinforce before measurement. Every algorithm in Weeks 11–15 is an instance of that pattern, and it is why quantum computers are not general-purpose accelerators.

Analogy boundary — the qubit as a probabilistic bit. A qubit is not a biased coin. A biased coin is described by one real number $p \in [0,1]$; a qubit needs two real parameters, and the extra one — relative phase — is invisible to any single measurement yet determines the result of measurements in other bases. Model a qubit as a random bit and every interference effect disappears from your predictions, which means every quantum algorithm becomes inexplicable.

Reversibility

Unitary operators satisfy $U^\dagger U = I$, so every quantum gate is invertible. This is not a design choice; it follows from the Schrödinger equation, and it has a physical basis.

Landauer's principle, 1961: erasing one bit dissipates at least $kT \ln 2$ of energy. Erasure is the only fundamentally irreversible operation, and it costs. Bennett showed in 1973 that any computation can be made reversible by keeping enough intermediate state.

Practical consequence: classical AND is irreversible — output 0 does not determine the inputs. So there is no two-qubit "AND gate." The quantum version is **Toffoli** (CCNOT), which is three-qubit and reversible: it flips the target if both controls are 1, keeping the controls. Reversible computation must carry its garbage, and managing that is a real cost in circuit design.

Analogy boundary — reversibility as an append-only log. Your hash chain is append-only, and reconstructing prior state from the log is a genuine analogue of unitary invertibility. The boundary is *where the information sits*. Your log grows — reversibility is purchased with storage. A unitary keeps the state space fixed and rearranges it; no records accumulate. Quantum computation is reversible without a log, which is why uncomputation (running a subroutine backwards to clear ancillas) is mandatory rather than optional.

The no-cloning theorem

Claim. No unitary U satisfies $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$.

Proof. Suppose it did, for $|\psi\rangle$ and $|\phi\rangle$. Unitaries preserve inner products:

$$\langle\psi|\psi\rangle = \langle\psi|\psi\rangle \cdot \langle\psi|\psi\rangle \quad \Rightarrow \quad z = z^2 \quad \Rightarrow \quad z \in \{0, 1\}$$

So cloning works only for states that are identical or orthogonal — that is, for a pre-agreed basis, which is classical copying. Arbitrary unknown states cannot be duplicated. The proof is four lines and follows from linearity alone. ■

Consequences:

- **No quantum backups.** You cannot snapshot a register mid-computation.
- **No copy-based error correction.** Classical repetition codes are unavailable; quantum error correction (Week 16) spreads one logical qubit across many physical ones *without* ever copying the state.

- **Quantum key distribution is secure by physics.** An eavesdropper cannot copy the transmitted qubits and must measure, which disturbs them detectably. BB84 rests entirely on this.

Analogy boundary — no-cloning as tamper evidence. This is the strongest analogy available to you, and it is nearly exact in its consequences: an unknown quantum state cannot be duplicated without detection, exactly as a hash-chained record cannot be altered without breaking the chain. Both make interception evident rather than impossible. The boundary is the source of the guarantee. Your chain is secure because inverting SHA-256 is *computationally* infeasible — a sufficiently powerful adversary, or a broken hash, defeats it. No-cloning is **information-theoretic**: it holds against unlimited computation, forever, because it follows from linearity. That distinction is the entire argument for QKD over classical key exchange, and it is worth being precise about in any threat model you write.

Key takeaways

- Tunneling probability falls exponentially with barrier width and $\sqrt{m}$ — enabling STM and flash memory, and setting the leakage floor that ended Dennard scaling.
- n qubits carry 2^n amplitudes but yield only n bits on measurement. Exponential description, linear readout.
- Quantum advantage comes from engineered interference, not parallelism.
- All quantum gates are unitary and reversible; irreversible classical gates like AND require reversible embeddings such as Toffoli.
- No-cloning follows from linearity in four lines. It forbids quantum backups and repetition codes, and it makes QKD information-theoretically secure.

Self-check

Q1 (Objective 2) — You are asked to design a provenance system where quantum states are signed and forwarded across custody boundaries. Explain why a straightforward port of your hash-chain design fails, and identify which property you must give up.

Q2 (Objective 2) — A vendor claims their 60-qubit machine "evaluates 2^{60} possibilities simultaneously" and therefore breaks AES-128. Give the two independent reasons this is wrong.

Check your answers

A1 — The port fails at the first step: **hashing requires reading the input**, and reading a quantum state destroys it. Your design assumes you can compute a digest over a record while leaving the record intact — a non-destructive read. For an unknown quantum state, no such operation exists. Measuring to compute a hash collapses the state, so what you forward is a classical outcome, not the state you meant to sign.

Nor can you clone first and hash the copy: no-cloning forbids it, and it forbids it for exactly the states you care about (unknown, arbitrary ones).

What you must give up is **the digest-of-content model**. Viable alternatives shift the guarantee elsewhere:

- Sign *classical metadata* — preparation procedure, timestamps, device identity — and accept that the quantum payload is authenticated by provenance of process, not of content.
- Use **quantum authentication** schemes, which encode the state so tampering is detectable with high probability, at the cost of ancillary qubits and a pre-shared key.
- Perform **destructive verification** on a random subsample, accepting statistical rather than per-record assurance — the same structure as BB84's parameter estimation.

The general lesson: your chain's integrity guarantee rests on cheap, repeatable, non-destructive reads. Remove that primitive and the architecture must change, not merely be tuned.

A2 — First: readout. The 2^{60} amplitudes exist in the description, but measurement returns one 60-bit string sampled from $|\text{amplitude}|^2$. Without an interference pattern that concentrates amplitude on the answer, you get a uniformly random string — precisely as useful as guessing. "Evaluates simultaneously" conflates the state's dimension with accessible output.

Second: the actual quantum bound on AES. Grover's algorithm (Week 13) gives a *quadratic* speedup for unstructured search, reducing 2^{128} operations to $\sim 2^{64}$ — not to 1. That remains computationally infeasible, and it is why the standard guidance is to double symmetric key lengths (AES-256 $\rightarrow$ $\sim 2^{128}$ effective) rather than abandon symmetric cryptography. AES has no structure Shor's algorithm can exploit; only the asymmetric primitives (RSA, ECC) face exponential attack.

A third point worth adding: a 60-qubit NISQ device without error correction cannot sustain coherence through the $\sim 2^{64}$ sequential Grover iterations anyway. The gap between qubit count and useful logical operations is the subject of Week 16, and it is the number that actually matters in a threat model.